

## VISHAKHAPATNAM CWC, India

WMO: 431500

Lat: 17.7217N

Lon: 83.3292E

Elev: 66

StdP: 100.53

Time Zone: 5.50 (IND)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.1       | 20.8 | 11.0                          | 8.2 | 25.1 | 12.9 | 9.3 | 24.6 | 7.3                   | 24.0 | 6.4 | 24.5 | 1.3                   | 0    | 0.345 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 4.1                    | 34.0            | 27.1 | 33.2 | 27.5 | 32.7 | 27.5 | 29.3                | 32.3 | 28.9 | 31.9 | 28.5 | 31.6 | 2.9                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 28.5                            | 25.1 | 31.7 | 28.1 | 24.4 | 31.4 | 27.6 | 23.7 | 31.1 | 97.2          | 32.4 | 95.1 | 32.0 | 92.9 | 31.6 | 30.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|------|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |      |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |      |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)  | (q)  |
| 7.4               | 6.3  | 5.4 | DB                         | 18.1 | 37.5               | 1.5 | 1.5                                                | 17.1 | 38.6       | 16.2 | 39.5       | 15.3 | 40.3       | 14.3 | 41.4 | 41.4 |
|                   |      |     | WB                         | 13.9 | 29.9               | 1.2 | 0.2                                                | 13.0 | 30.0       | 12.3 | 30.2       | 11.6 | 30.3       | 10.7 | 30.5 | 30.5 |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 27.9   | 24.4  | 25.9  | 27.9  | 29.3  | 30.5  | 30.0  | 28.9  | 28.7  | 28.7  | 28.4  | 26.7  | 24.9 |      |
|                                                                               | DBStd                                                                         | 2.22   | 0.99  | 1.08  | 0.92  | 0.97  | 1.23  | 1.64  | 1.46  | 1.16  | 1.08  | 1.15  | 1.14  | 1.17 |      |
|                                                                               | HDD10.0                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | HDD18.3                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | CDD10.0                                                                       | 6522   | 446   | 444   | 556   | 578   | 637   | 600   | 585   | 578   | 562   | 570   | 502   | 462  |      |
|                                                                               | CDD18.3                                                                       | 3480   | 188   | 211   | 298   | 328   | 378   | 350   | 327   | 320   | 312   | 312   | 252   | 204  |      |
|                                                                               | CDH23.3                                                                       | 39999  | 1061  | 1751  | 3423  | 4308  | 5388  | 4783  | 4111  | 3935  | 3845  | 3708  | 2373  | 1314 |      |
|                                                                               | CDH26.7                                                                       | 15823  | 111   | 357   | 1143  | 1946  | 2924  | 2418  | 1692  | 1531  | 1517  | 1384  | 606   | 196  |      |
| Wind                                                                          | WSAvg                                                                         | 2.3    | 1.7   | 1.8   | 2.3   | 2.9   | 2.7   | 2.6   | 2.3   | 2.2   | 1.7   | 2.1   | 2.6   | 2.2  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 996    | 9     | 12    | 12    | 30    | 47    | 105   | 131   | 155   | 172   | 225   | 86    | 7    |      |
|                                                                               | PrecMax                                                                       | 1451   | 88    | 114   | 194   | 276   | 211   | 341   | 355   | 430   | 405   | 668   | 367   | 76   |      |
|                                                                               | PrecMin                                                                       | 641    | 0     | 0     | 0     | 0     | 4     | 17    | 28    | 27    | 36    | 12    | 0     | 0    |      |
|                                                                               | PrecStd                                                                       | 210    | 16    | 26    | 34    | 48    | 40    | 81    | 67    | 74    | 82    | 158   | 103   | 16   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 29.1  | 30.9  | 32.1  | 33.1  | 35.3  | 35.5  | 34.7  | 33.5  | 33.2  | 33.1  | 31.6  | 30.1 |      |
|                                                                               |                                                                               | MCWB   | 22.4  | 23.2  | 24.6  | 27.4  | 27.2  | 27.1  | 26.0  | 25.9  | 26.9  | 25.5  | 22.8  | 22.6 |      |
|                                                                               | 2%                                                                            | DB     | 28.4  | 30.0  | 31.2  | 32.2  | 33.7  | 34.0  | 32.7  | 32.1  | 32.2  | 32.2  | 30.8  | 29.2 |      |
|                                                                               |                                                                               | MCWB   | 22.3  | 23.6  | 25.4  | 27.1  | 27.9  | 27.6  | 26.8  | 26.8  | 27.2  | 25.9  | 23.7  | 22.2 |      |
|                                                                               | 5%                                                                            | DB     | 27.9  | 29.3  | 30.8  | 31.7  | 33.1  | 33.1  | 31.9  | 31.4  | 31.7  | 31.7  | 30.2  | 28.5 |      |
|                                                                               |                                                                               | MCWB   | 22.1  | 23.5  | 25.4  | 26.9  | 28.2  | 27.7  | 26.9  | 26.8  | 27.1  | 25.9  | 23.6  | 21.9 |      |
|                                                                               | 10%                                                                           | DB     | 27.3  | 28.7  | 30.3  | 31.3  | 32.7  | 32.4  | 31.1  | 30.8  | 31.1  | 31.0  | 29.5  | 27.8 |      |
|                                                                               |                                                                               | MCWB   | 21.7  | 23.3  | 25.3  | 26.8  | 28.1  | 27.6  | 26.8  | 26.8  | 27.0  | 25.9  | 23.4  | 21.5 |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 24.6  | 26.0  | 27.1  | 28.6  | 29.8  | 29.6  | 28.7  | 28.2  | 28.5  | 28.1  | 26.6 | 24.9 |
|                                                                               |                                                                               |        | MCDB  | 27.3  | 29.2  | 30.4  | 31.7  | 32.7  | 32.9  | 31.7  | 30.9  | 31.3  | 31.1  | 29.5 | 28.3 |
| 2%                                                                            |                                                                               | WB     | 24.0  | 25.2  | 26.7  | 28.0  | 29.5  | 29.1  | 28.1  | 27.9  | 28.0  | 27.5  | 25.9  | 24.1 |      |
|                                                                               |                                                                               | MCDB   | 27.0  | 28.4  | 30.0  | 31.2  | 32.4  | 32.3  | 31.2  | 30.7  | 31.1  | 30.7  | 28.9  | 27.4 |      |
| 5%                                                                            |                                                                               | WB     | 23.3  | 24.7  | 26.3  | 27.6  | 29.1  | 28.7  | 27.7  | 27.6  | 27.7  | 27.1  | 25.4  | 23.5 |      |
|                                                                               |                                                                               | MCDB   | 26.5  | 28.0  | 29.6  | 30.9  | 32.1  | 31.9  | 30.7  | 30.4  | 30.7  | 30.4  | 28.4  | 26.8 |      |
| 10%                                                                           |                                                                               | WB     | 22.8  | 24.2  | 26.0  | 27.3  | 28.7  | 28.2  | 27.2  | 27.2  | 27.3  | 26.7  | 24.9  | 23.0 |      |
|                                                                               |                                                                               | MCDB   | 26.0  | 27.5  | 29.3  | 30.6  | 31.8  | 31.4  | 30.2  | 30.0  | 30.3  | 29.9  | 28.0  | 26.3 |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 6.2   | 6.0   | 5.2   | 4.3   | 4.1   | 3.9   | 3.7   | 4.0   | 4.4   | 5.0   | 5.8   | 6.3  |      |
|                                                                               |                                                                               | MCDBR  | 6.5   | 6.3   | 5.4   | 4.5   | 4.6   | 4.8   | 4.8   | 4.9   | 5.3   | 5.8   | 6.3   | 6.6  |      |
|                                                                               |                                                                               | MCWBR  | 2.9   | 3.2   | 2.8   | 2.7   | 3.5   | 3.3   | 2.8   | 2.7   | 2.8   | 2.6   | 2.8   | 3.0  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 5.5   | 5.4   | 4.8   | 4.0   | 4.0   | 4.0   | 4.1   | 4.2   | 4.6   | 4.9   | 5.1   | 5.5  |      |
|                                                                               |                                                                               | MCWBR  | 2.8   | 3.0   | 2.7   | 2.8   | 3.6   | 3.5   | 2.9   | 2.9   | 2.9   | 2.7   | 2.6   | 2.9  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.512  | 0.507 | 0.581 | 0.586 | 0.635 | 0.635 | 0.567 | 0.545 | 0.549 | 0.544 | 0.518 | 0.534 |      |      |
|                                                                               | taud                                                                          | 1.984  | 2.030 | 1.874 | 1.918 | 1.815 | 1.823 | 2.005 | 2.071 | 2.057 | 2.046 | 2.036 | 1.946 |      |      |
|                                                                               | Ebn at Noon                                                                   | 767    | 800   | 754   | 752   | 707   | 700   | 750   | 773   | 769   | 758   | 756   | 731   |      |      |
|                                                                               | Edh at Noon                                                                   | 171    | 171   | 206   | 198   | 217   | 213   | 178   | 168   | 169   | 166   | 161   | 173   |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 4.70   | 5.58  | 6.13  | 6.58  | 6.10  | 4.56  | 4.19  | 4.31  | 4.47  | 4.63  | 4.49  | 4.28  |      |      |
|                                                                               | RadStd                                                                        | 0.20   | 0.23  | 0.27  | 0.23  | 0.26  | 0.44  | 0.49  | 0.38  | 0.35  | 0.47  | 0.38  | 0.28  |      |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | +0.30 | N/A     | N/A    | +0.38   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) | +0.30 | N/A     | N/A    | +0.38   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page